

Day 1 - Database Fundamentals

1. Data, Information and Database

Term	Meaning	Example
Data	Raw facts with no meaning on their own	101 , Rahul , 78
Information	Data with meaning	Student 101, Rahul, scored 78
Database	Organised collection of data you can store and retrieve	The whole student table

A **database** stores data in a structured way so it can be found, updated and protected — reliably, and by many people at once.

2. Why Not Files or Excel?

Problem	What goes wrong
Searching	You must read the whole file to find one record
Duplication	The same record typed twice, in two files
No validation	Nothing stops marks of 500 , or a name in the marks column
No relationships	You cannot link a student to their course
Multi-user	Two people editing at once overwrite each other
Security	Anyone who opens the file sees everything
Crash safety	Power cut while saving = corrupted file

3. DBMS and RDBMS

DBMS = the software that manages the database. You never touch the data files — you ask the DBMS: You --SQL--> DBMS --> data files on disk .

RDBMS = a **relational** DBMS: it stores data in **tables** that can be **related** to each other.

	DBMS	RDBMS
Stores data as	Files / one structure	Tables (rows & columns)
Relationships	Not supported	Supported, using keys
Redundancy	High	Low (normalisation)
Language	Varies	SQL
Examples	Older file systems	MySQL, PostgreSQL, SQLite, Oracle

4. Tables, Rows, Columns and Keys

A **table** holds one kind of thing — students in one table, courses in another.

id	name	marks	
101	Rahul Verma	78	← one row = one record
102	Anita Sharma	95	

Row = one record · **Column** = one attribute · **Field** = one cell where a row and column meet.

Key	Meaning
Primary key	Uniquely identifies each row. Never repeats, never empty
Foreign key	Points to the primary key of another table
Composite key	A primary key made of two or more columns together
Unique key	Must be unique, <i>can</i> be empty, and a table can have many

Choose a primary key that never changes — a roll number, not a phone number.

5. What is SQL?

SQL = **Structured Query Language**, the standard language for relational databases. It is **declarative** — you say *what* you want, not *how* to get it.

```
SELECT name FROM students WHERE marks > 50;
```

- Keywords are **not case sensitive** (`SELECT` = `select`), but CAPITALS are the normal style.
- Every statement ends with a **semicolon** `;`.
- SQL is a standard (ANSI SQL); each database adds small extras.

6. The Five Families of SQL Commands

Family	Full name	Commands	Purpose
DDL	Data D efinition Language	<code>CREATE</code> , <code>ALTER</code> , <code>DROP</code> , <code>TRUNCATE</code>	Build/change the <i>structure</i>
DML	Data M anipulation Language	<code>INSERT</code> , <code>UPDATE</code> , <code>DELETE</code>	Change the <i>data</i>
DQL	Data Q uery Language	<code>SELECT</code>	Read the data
TCL	T ransaction Control Language	<code>COMMIT</code> , <code>ROLLBACK</code> , <code>SAVEPOINT</code>	Confirm or undo changes
DCL	Data C ontrol Language	<code>GRANT</code> , <code>REVOKE</code>	Permissions

DDL builds the box, DML fills the box, DQL reads the box.

7. How a Query Runs

Your SQL → 1. **PARSER** (syntax, tables exist?) → 2. **OPTIMISER** (fastest plan, use an index?) → 3. **EXECUTION PLAN** → 4. **ENGINE** (rows read from disk) → 5. **RESULT**

The **optimiser** is why SQL is declarative: two people can write the same question differently and get the same plan.

Written order vs run order — you write `SELECT ... FROM ... WHERE ... GROUP BY ... HAVING ... ORDER BY ... LIMIT` , but it runs:

`FROM` (1) → `WHERE` (2) → `GROUP BY` (3) → `HAVING` (4) → `SELECT` (5) → `ORDER BY` (6) → `LIMIT` (7)

`FROM` runs **first** and `SELECT` almost **last** — which is why a `SELECT` alias cannot be used inside `WHERE` .

8. MySQL, SQLite, PostgreSQL

SQL is the language; MySQL is one program that speaks it — like British and American English.

	MySQL	SQLite	PostgreSQL
Type	Client–server	Serverless , one file	Client–server
Setup	Install server, set password	Just a file	Install server, create role
Port	3306	none	5432
Used for	Most web applications	Phones, browsers, tests	Analytics, complex data
Typing	strict	flexible	strictest

About **90% transfers directly** — `SELECT` , `WHERE` , `JOIN` , `GROUP BY` , `HAVING` , subqueries, window functions are identical. The differences are at the edges:

Job	MySQL	SQLite / PostgreSQL
Join two strings	<code>CONCAT(a, b)</code>	<code>a b</code>
Today's date	<code>CURDATE()</code>	<code>DATE('now')</code> / <code>CURRENT_DATE</code>
Year from a date	<code>YEAR(d)</code>	<code>STRFTIME('%Y', d)</code> / <code>EXTRACT(...)</code>

9. Installing MySQL and Using Workbench

Download the **MySQL Installer for Windows** from <https://dev.mysql.com/downloads/installer/> (the larger *community* file), choose **Developer Default** — it installs MySQL Server, Workbench, Shell and connectors — keep the port **3306**, set a **root password** at *Accounts and Roles* and write it down, leave *Configure MySQL Server as a Windows Service* ticked, then open Workbench and click **Local instance MySQL80**. Lighter alternatives: **XAMPP** (MySQL + phpMyAdmin) or **Docker** — `docker run -d -p 3306:3306 -e MYSQL_ROOT_PASSWORD=root mysql:8` .

Workbench: **Navigator** (left) lists databases and tables, the **Query tab** is where you type SQL, the **Result grid** shows the rows, the **Output panel** shows messages. Run SQL with **Ctrl + Enter** (current statement) or **Ctrl + Shift + Enter** (whole tab). MySQL **saves as you go** — there is no "write changes" button, unless you open a transaction yourself. Workbench's **safe update mode** refuses `UPDATE` / `DELETE` without a key in `WHERE` (*Error 1175*); turn it off with `SET SQL_SAFE_UPDATES = 0;` .

10. Databases Inside the Server, and Your First SQL

One MySQL server holds **many databases**, so you must say which one you mean. Forget `USE` and you get "No database selected".

```

CREATE DATABASE IF NOT EXISTS training;  -- DDL
USE training;
SHOW DATABASES;
SHOW TABLES;

CREATE TABLE students (                -- DDL
    id    INT PRIMARY KEY,
    name  VARCHAR(50) NOT NULL,
    marks INT
);

INSERT INTO students (id, name, marks) VALUES  -- DML
    (101, 'Rahul Verma', 78),
    (102, 'Anita Sharma', 95),
    (103, 'Karan Patel', 38);

SELECT * FROM students;                -- DQL
SELECT name, marks FROM students WHERE marks > 50 ORDER BY marks DESC;
DESCRIBE students;

```

```

DESCRIBE students;

```

Field	Type	Null	Key	Default
id	int	NO	PRI	NULL
name	varchar(50)	NO		NULL
marks	int	YES		NULL

11. Common Mistakes

1. **Forgetting** `USE training;` — "No database selected".
2. **Forgetting the semicolon** — Workbench waits for you to finish the statement.
3. **Using** `||` **to join strings** — in MySQL that means **OR** and returns `0`. Use `CONCAT(a, b)`.
4. **Double quotes for text** — use **single quotes**: `'Rahul'` (PostgreSQL rejects `"Rahul"`).
5. **= vs ==** — SQL comparison is a **single** `=`.
6. **Error 1175 on UPDATE / DELETE** — safe update mode; add a `WHERE` or `SET SQL_SAFE_UPDATES = 0;`.
7. **Losing the root password** — there is no easy way back.

12. Summary

- A **database** solves searching, duplication, validation, relationships, multi-user access and crash safety.
- A **DBMS** manages the database; an **RDBMS** stores it as **related tables**.
- Data lives in **tables** of **rows** and **columns**, identified by a **primary key**, linked by a **foreign key**.
- **SQL** is declarative, in five families: **DDL, DML, DQL, TCL, DCL**.
- A query is **parsed** → **optimised** → **planned** → **executed**; `FROM` runs first, `SELECT` almost last.
- **SQL is the language; MySQL is one engine** — ~90% transfers to PostgreSQL, SQLite and Oracle.
- MySQL is a **server** holding many databases, so always `USE` one.

13. Practice Questions

1. Give three problems with storing 50,000 records in Excel that a database solves.
2. What is the difference between a DBMS and an RDBMS? And between a primary key and a unique key?
3. Which family does each belong to — `CREATE`, `INSERT`, `SELECT`, `ROLLBACK`?
4. Why can a `SELECT` alias not be used inside `WHERE`?
5. In one sentence each: what is SQL, and what is MySQL? Name two differences between MySQL and SQLite.
6. Why does `SELECT 'a' || 'b';` return `0` in MySQL?
7. Install MySQL, connect with Workbench, create the database `training` and select it.
8. Create a `courses` table with `course_id`, `course_name` and `duration`; insert three rows and select those longer than 30 days.
9. Run `DESCRIBE courses;` and explain each column of the output.
10. Insert a course with a duplicate `course_id` and read the error carefully.

Day 2 - Creating, Inserting, Altering and Deleting

1. Working with Databases

A database is an organised collection of related data. It holds one or more tables, and every table belongs to a database — so the database must exist and be selected before you create a table.

Command	Purpose
CREATE DATABASE db;	Creates a new database
CREATE DATABASE IF NOT EXISTS db;	Creates it only if it does not exist
SHOW DATABASES;	Lists all databases on the server
USE db;	Selects the database for this session
SELECT DATABASE();	Shows the currently selected database
DROP DATABASE db;	Deletes the database and everything in it
DROP DATABASE IF EXISTS db;	Deletes it only if it exists

```
CREATE DATABASE IF NOT EXISTS training;
USE training;                -- "Database changed"
SELECT DATABASE();           -- training
```

DROP DATABASE cannot be undone.

Error	Meaning	Fix
1007 Can't create database; database exists	Name already taken	CREATE DATABASE IF NOT EXISTS ...
1049 Unknown database	It does not exist	SHOW DATABASES; then USE an existing one

2. Tables, Data Types and Constraints

A **table** stores related data as **rows** (records) and **columns** (fields). Each column has a name, a data type and optional constraints, and each table name must be unique inside its database.

Data type	Use	Example
INT / BIGINT	Whole numbers	100 / 9999999999
FLOAT / DOUBLE	Decimal numbers	89.75 / 12345.6789
DECIMAL(10,2)	Fixed decimals (money)	1999.99
CHAR(n) / VARCHAR(n)	Fixed / variable length text	'A' / 'Python'
TEXT	Large text	a description
DATE / TIME / DATETIME	Date / time / both	'YYYY-MM-DD HH:MM:SS'
BOOLEAN	TRUE or FALSE	TRUE

Constraint	Purpose	Example
PRIMARY KEY	Uniquely identifies each record	course_id
NOT NULL	Prevents NULL values	course_name
UNIQUE	Prevents duplicate values	email
DEFAULT	Assigns a default value	fee DEFAULT 5000
CHECK	Restricts allowed values	fee > 0
AUTO_INCREMENT	Generates values automatically	student_id
FOREIGN KEY	Links two tables	department_id

3. Creating Tables

```
CREATE TABLE courses (                                -- 1. basic
  course_id  INT PRIMARY KEY,
  course_name VARCHAR(50) NOT NULL,
  duration   INT,
  fee        INT DEFAULT 5000
);

CREATE TABLE IF NOT EXISTS courses (...);    -- 2. only if it does not exist

CREATE TABLE courses_backup AS SELECT * FROM courses;      -- 3. structure + data
CREATE TABLE courses_empty AS SELECT * FROM courses WHERE 1 = 0; -- 4. structure only
CREATE TABLE course_catalog AS SELECT course_id, course_name FROM courses; -- 5. chosen columns
CREATE TABLE premium_courses AS SELECT * FROM courses WHERE fee >= 7000; -- 6. chosen rows

CREATE TABLE courses_clone LIKE courses;    -- 7. full structure, no data

CREATE TEMPORARY TABLE temp_courses (      -- 8. dropped when the session ends
  course_id INT, course_name VARCHAR(50)
);

CREATE TABLE trainers (                    -- 9. auto-generated ids
  trainer_id  INT AUTO_INCREMENT PRIMARY KEY,
  trainer_name VARCHAR(50),
  experience   INT
);
```

LIKE copies column names, data types, the primary key, indexes and AUTO_INCREMENT — but **not** the data. AS SELECT copies data but not keys and indexes.

Error	Meaning	Fix
1050 Table already exists	Name in use	CREATE TABLE IF NOT EXISTS ...
1046 No database selected	No USE yet	USE training;
1060 Duplicate column name	Column listed twice	Rename one column
1064 Syntax error	Missing brackets or commas	Wrap columns in (...) , comma between them

4. Looking at Tables

```
SHOW TABLES;      -- all tables in the selected database
DESCRIBE courses;   -- or DESC courses;
SHOW CREATE TABLE courses; -- the exact CREATE statement
```

Field	Type	Null	Key	Default	Extra
course_id	int	NO	PRI	NULL	
course_name	varchar(50)	NO		NULL	
duration	int	YES		NULL	
fee	int	YES		5000	

5. Inserting Data

Values must match the column data types and the column order, text goes in **single quotes**, and an AUTO_INCREMENT column is filled in for you.

```

INSERT INTO courses VALUES (101, 'Python', 30, 5000);           -- 1. one record

INSERT INTO courses VALUES                                     -- 2. many records
(102, 'Java', 45, 7000),
(103, 'SQL', 20, 3000),
(104, 'Web Development', 60, 9000);

INSERT INTO trainers(trainer_name, experience)                 -- 3. chosen columns
VALUES ('Rahul', 5);

INSERT INTO trainers(trainer_name, experience)                 -- 4. many, chosen columns
VALUES ('Anita', 8), ('Kiran', 4), ('David', 6);

INSERT INTO courses_backup SELECT * FROM courses;              -- 5. from another table

INSERT INTO course_catalog(course_id, course_name)             -- 6. chosen columns from another table
SELECT course_id, course_name FROM courses;

```

Error	Meaning	Reason
1062 Duplicate entry '101' for key 'PRIMARY'	Id already used	A primary key must be unique
1136 Column count doesn't match value count	Too few / too many values	List the columns, or give every value
1048 Column 'course_name' cannot be null	NULL into a NOT NULL column	Supply a value
1366 Incorrect integer value: 'ABC'	Text into an INT column	Match the data type

6. Changing a Table — ALTER TABLE

ALTER TABLE changes the **structure** only; the existing data stays.

Job	Statement
Add a column	ALTER TABLE courses ADD COLUMN trainer_name VARCHAR(50);
Add several	ALTER TABLE courses ADD COLUMN start_date DATE, ADD COLUMN mode VARCHAR(20);
Change a data type	ALTER TABLE courses MODIFY COLUMN course_name VARCHAR(100);
Rename a column	ALTER TABLE courses RENAME COLUMN fee TO course_fee;
Rename the table	ALTER TABLE courses RENAME TO course_details;
Drop a column	ALTER TABLE course_details DROP COLUMN trainer_name;
Move to first position	ALTER TABLE course_details MODIFY COLUMN course_fee INT FIRST;
Place after a column	ALTER TABLE course_details MODIFY COLUMN course_fee INT AFTER duration;
Set a default	ALTER TABLE course_details ALTER COLUMN mode SET DEFAULT 'Offline';
Remove a default	ALTER TABLE course_details ALTER COLUMN mode DROP DEFAULT;

Check the result with DESC course_details;. Older MySQL versions use MODIFY COLUMN mode VARCHAR(20) DEFAULT 'Offline' instead of ALTER COLUMN ... SET DEFAULT .

Error	Meaning	Fix
1091 Can't DROP 'salary'	No such column	DESC the table first
1146 Table doesn't exist	Wrong table name	SHOW TABLES;
1060 Duplicate column name	Column already there	Pick another name

7. Removing Data, Tables and Databases

Command	Removes data	Removes structure	Can use WHERE	Can be rolled back
DELETE	Yes	No	Yes	Yes
TRUNCATE	Yes	No	No	No
DROP TABLE	Yes	Yes	No	No

Rollback depends on the storage engine and transaction settings.

```
DELETE FROM courses WHERE course_id = 103; -- one record
DELETE FROM courses WHERE fee < 6000;      -- every matching record
DELETE FROM courses;                       -- all rows, table stays -> "Empty set"

TRUNCATE TABLE courses;                  -- all rows at once, faster, structure stays

DROP TABLE courses;                     -- table and data gone
DROP TABLE IF EXISTS courses;           -- no error if it is missing
DROP TABLE trainers, course_catalog;    -- several at once

DROP DATABASE training;                  -- database and every table inside it
DROP DATABASE IF EXISTS training;
```

Error	Meaning	Fix
1051 Unknown table 'students'	No such table	SHOW TABLES; or DROP TABLE IF EXISTS ...
1008 Can't drop database 'college'	No such database	SHOW DATABASES; or DROP DATABASE IF EXISTS ...

8. Summary

- Create and select a database first: CREATE DATABASE → USE → SELECT DATABASE() .
- A table is columns with a **data type** and optional **constraints** (PRIMARY KEY , NOT NULL , UNIQUE , DEFAULT , CHECK , AUTO_INCREMENT , FOREIGN KEY).
- Tables can be created from scratch, from a SELECT (data, structure, chosen columns or rows), with LIKE , or as TEMPORARY .
- Inspect with SHOW TABLES , DESC and SHOW CREATE TABLE .
- INSERT one row, many rows, chosen columns, or rows copied from another table.
- ALTER TABLE adds, modifies, renames, repositions and drops columns, and renames the table.
- DELETE removes chosen rows, TRUNCATE empties a table fast, DROP removes the table or database completely.

Next topic: **retrieving data with SELECT** .

9. Practice Questions

1. Create the database training , select it, and confirm with SELECT DATABASE(); .
2. Create a students table with an id, name, marks and a DEFAULT fee.
3. Insert one student, then five students in a single statement.
4. Create students_backup with the same structure but no data, then copy all records into it.
5. Create a table holding only student_id and student_name .
6. Create a clone of students using LIKE , and a temporary table.
7. Add a column category , then add rating and language together.
8. Change student_name from VARCHAR(50) to VARCHAR(150) .
9. Rename the column fee to course_fee , and the table students to student_details .
10. Move course_fee to the first position, then drop the category column, and check with DESC .
11. Delete one student by id, then all students with marks below 35.
12. Empty the table with TRUNCATE , prove the structure still exists, then drop the table and the database.